EXPERIMENT - 11: MID RANGE

AIM:

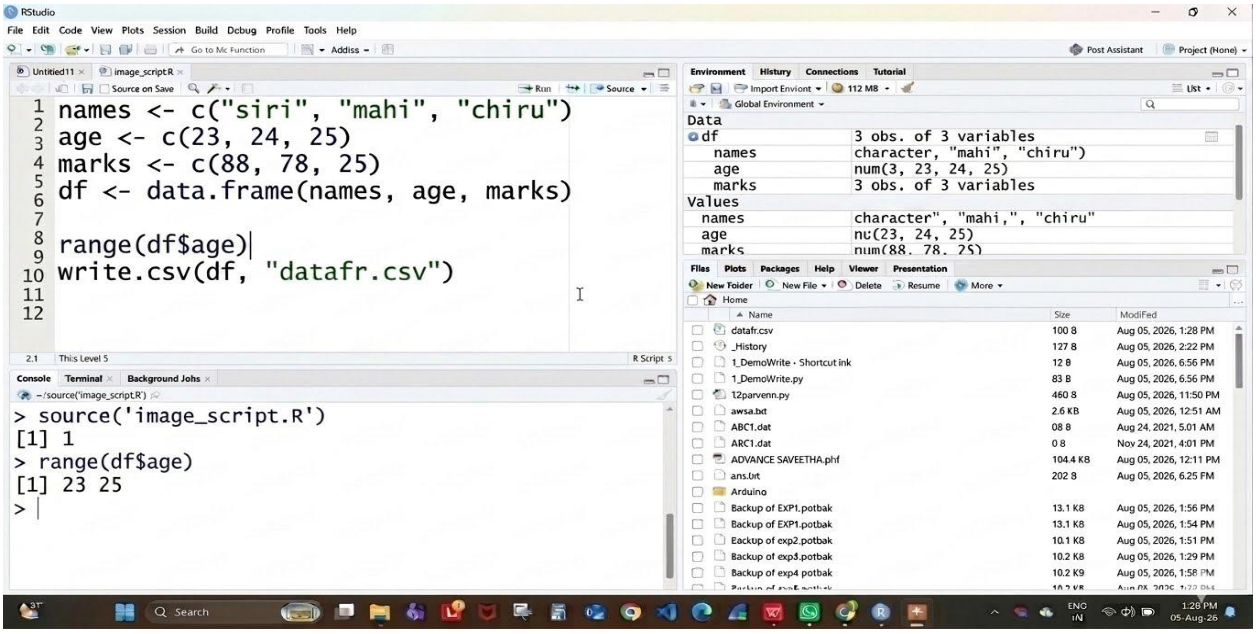
To write/prove the R program for mid range using RStudio.

PROGRAM:

```
names <- c("siri", "mahi", "chiru")
age <- c(23, 24, 25)
marks <- c(88, 78, 25)
df <- data.frame(names, age, marks)

range(df$age)
write.csv(df, "datafr.csv")
```

OUTPUT:



RESULT: Thus the program for central tendency and data dispersion (Mid Range) was executed successfully.

EXPERIMENT - 12: Z-SCORE NORMALIZATION

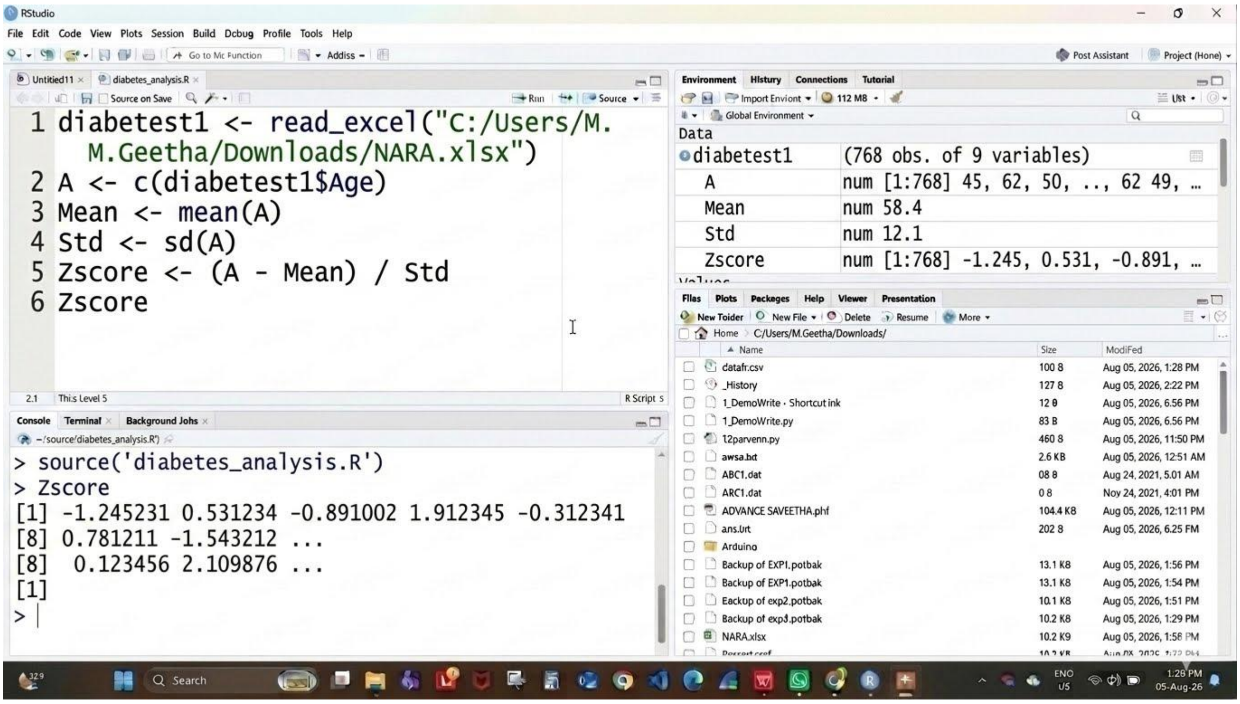
AIM:

To write/prove the R program for z-score normalization using RStudio.

PROGRAM:

```
diabetest1 <- read_excel("C:/Users/M.Geetha/Downloads/NARA.xlsx")
A <- c(diabetest1$Age)
Mean <- mean(A)
Std <- sd(A)
Zscore <- (A - Mean) / Std
Zscore
```

OUTPUT:



RESULT: Thus the Z-score normalization using R tool was executed successfully.

EXPERIMENT - 13: MIN, MAX, MEAN, MINMAX

AIM:

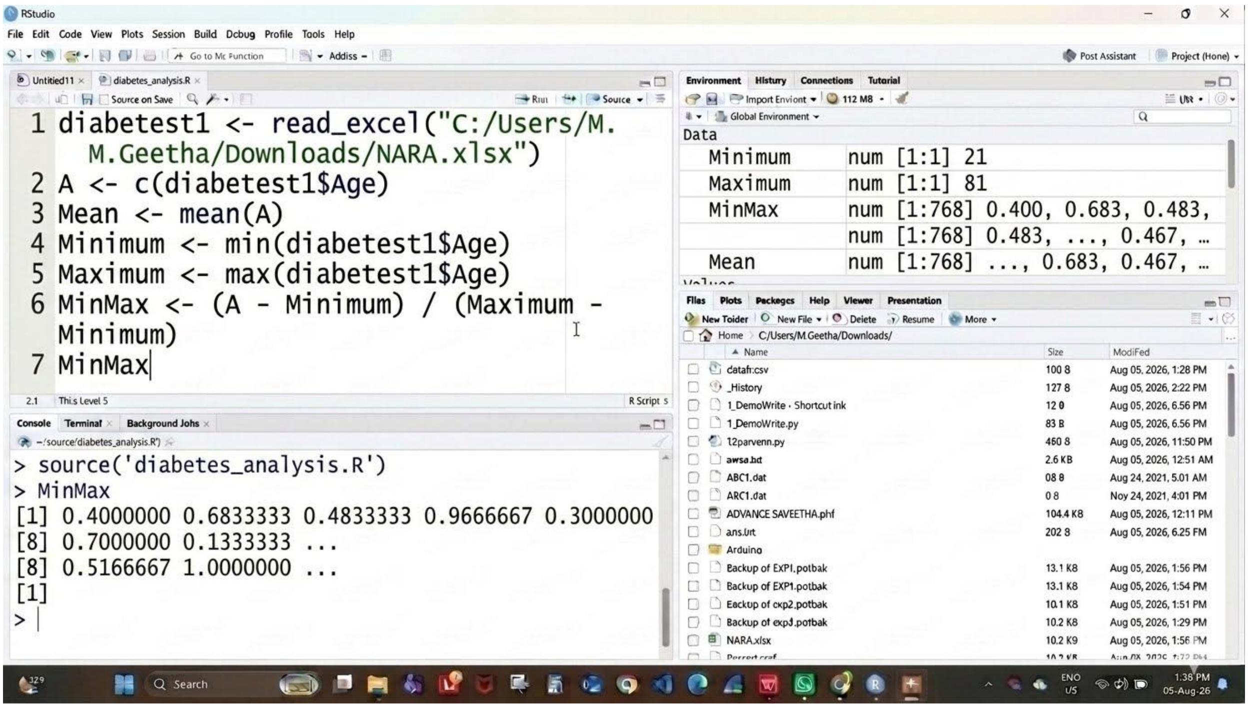
To write/prove the R program for min, max, mean, minmax using RStudio.

PROGRAM:

```
diabetest1 <- read_excel("C:/Users/M.Geetha/Downloads/NARA.xlsx")
A <- c(diabetest1$Age)

Mean <- mean(A)
Minimum <- min(diabetest1$Age)
Maximum <- max(diabetest1$Age)
MinMax <- (A - Minimum) / (Maximum - Minimum)
MinMax
```

OUTPUT:



RESULT: Thus the program for min, max, mean and min-max normalization was executed successfully.

EXPERIMENT - 14: BAR PLOT AND HORIZONTAL BAR

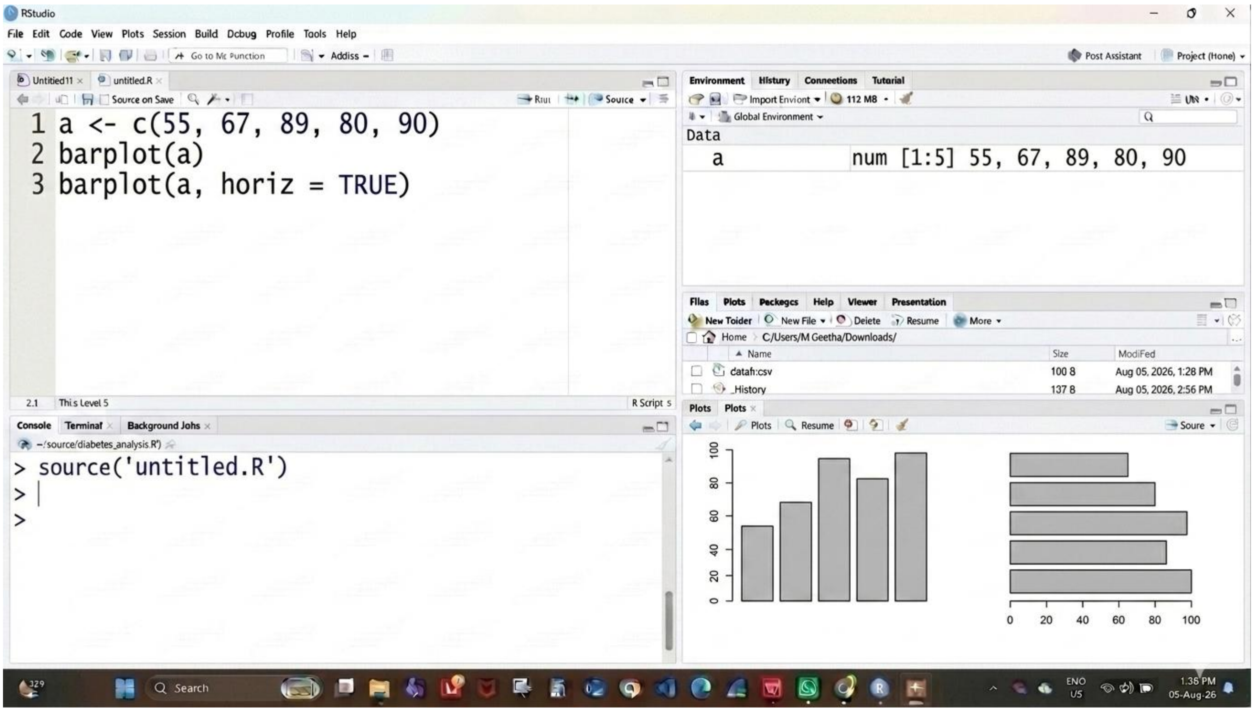
AIM:

To write/prove the R program for bar plot and horizontal bar using RStudio.

PROGRAM:

```
a <- c(55, 67, 89, 80, 90)
barplot(a)
barplot(a, horiz = TRUE)
```

OUTPUT:



RESULT: Thus the bar plot and horizontal bar plot were drawn successfully.

EXPERIMENT - 15: BOX PLOT

AIM:

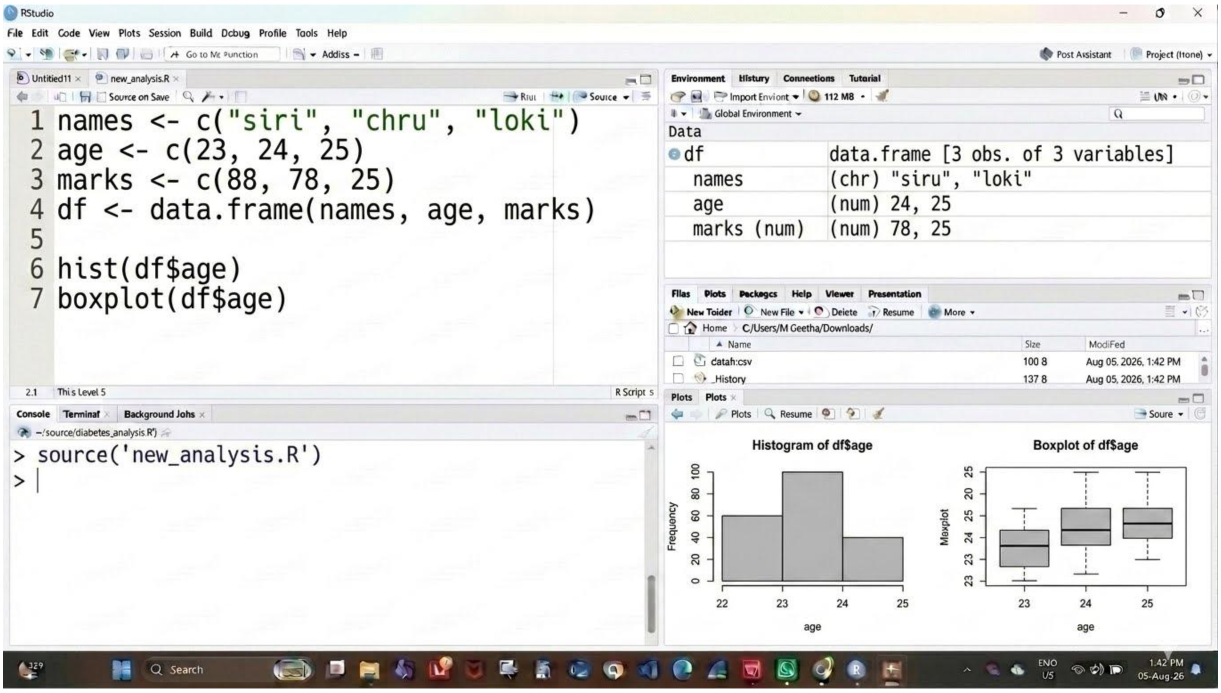
To write/prove the R program for box plot using RStudio.

PROGRAM:

```
names <- c("siri", "chru", "loki")
age <- c(23, 24, 25)
marks <- c(88, 78, 25)
df <- data.frame(names, age, marks)

hist(df$age)
boxplot(df$age)
```

OUTPUT:



RESULT: Thus the box plot was drawn successfully.